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Intelligent Data Processing and Applications

COE 543/743

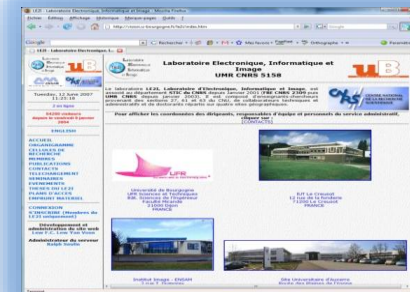
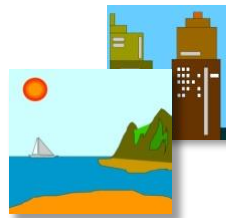
Chapter 4 - Overview of Similarity Processing

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Computer Software Engineering

Ch. 4 – Overview of Similarity Processing

- The semi-structured nature of data makes it ideal for representing heterogeneous data
 - Information** to be **broadcasted over the Web** is described in **semi-structured** and **NoSQL** data
 - ⇒ To guarantee its interoperability!

```
<Academy Name='Dijon'>
  <Department>
    <Laboratory>
      <Professor>John Cramer</Professor>
      <Student>John Takagi</Student>
    </Laboratory>
  </Department>
</Academy>
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Eminent need in DB, IR, and DM

Need for semi-structured data similarity evaluation

↳ **Toward Intelligent Data Processing on the Web**

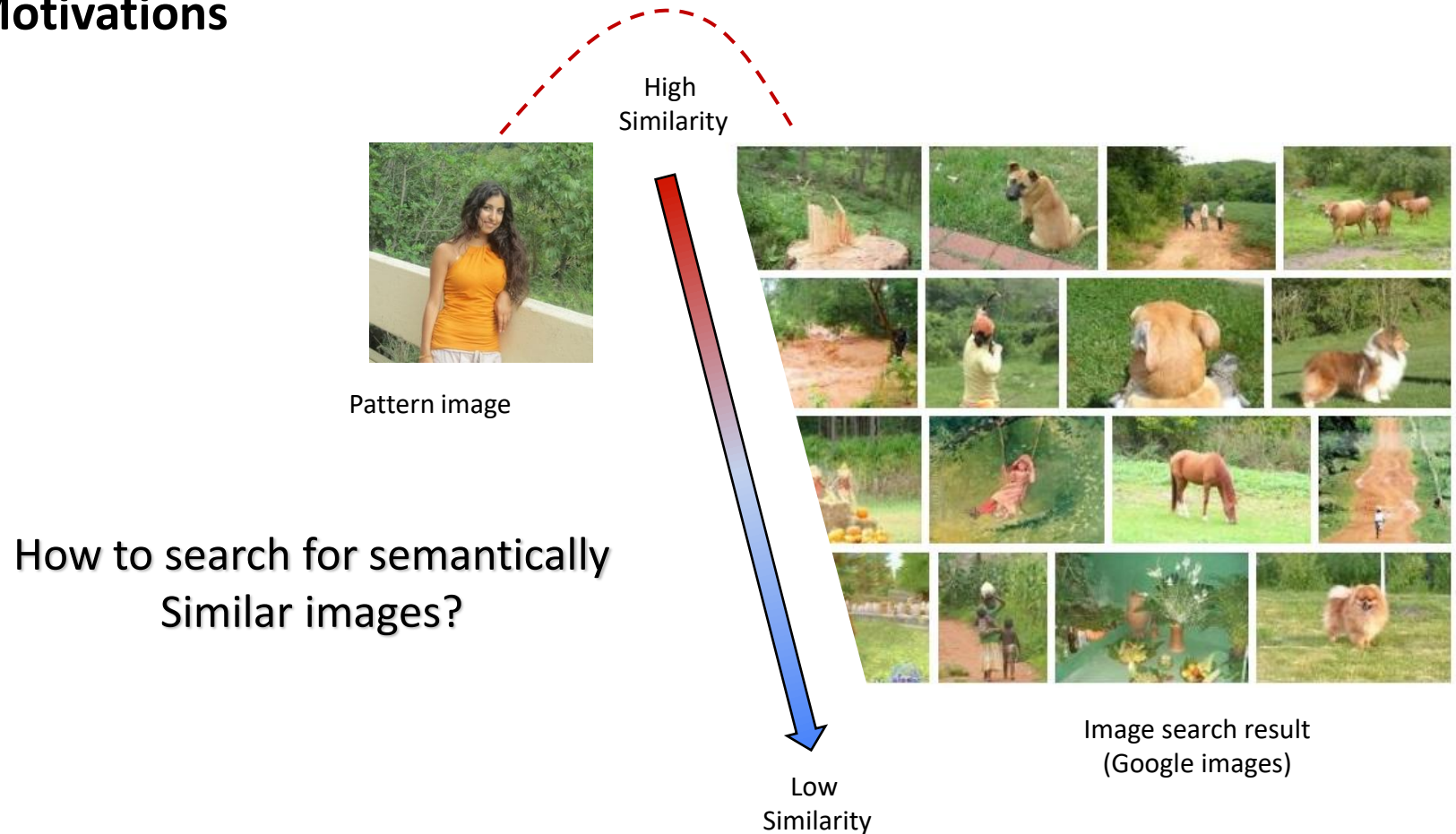
Ch. 4 – Overview of Similarity Processing

Chapter Contents

1. Motivations
2. Formal Definition of Similarity
3. Main Categories of Similarity Approaches
4. Edit Distance (ED) based Approaches
5. Information Retrieval (IR) based Approaches
6. Other Approaches
7. Current Issues
8. Ongoing Solutions
9. Toward an Intelligent Web

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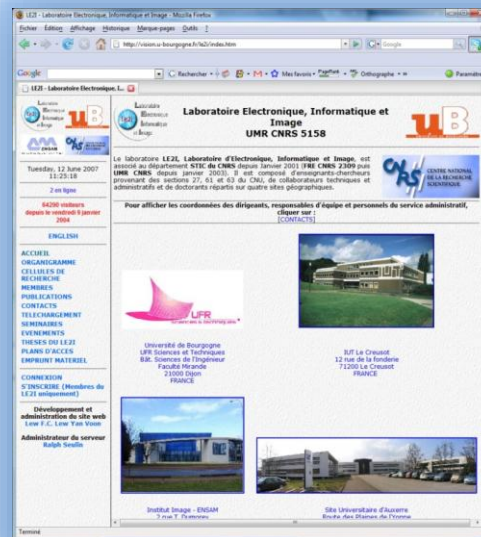
1. Motivations



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1. Motivations

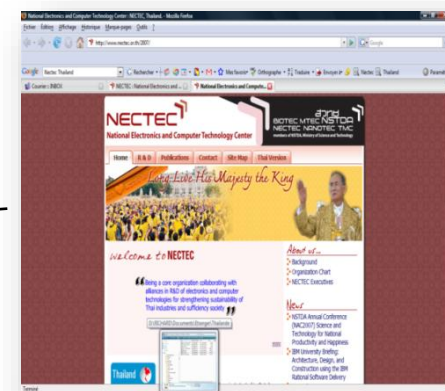
How to merge
Web documents?



High
Similarity



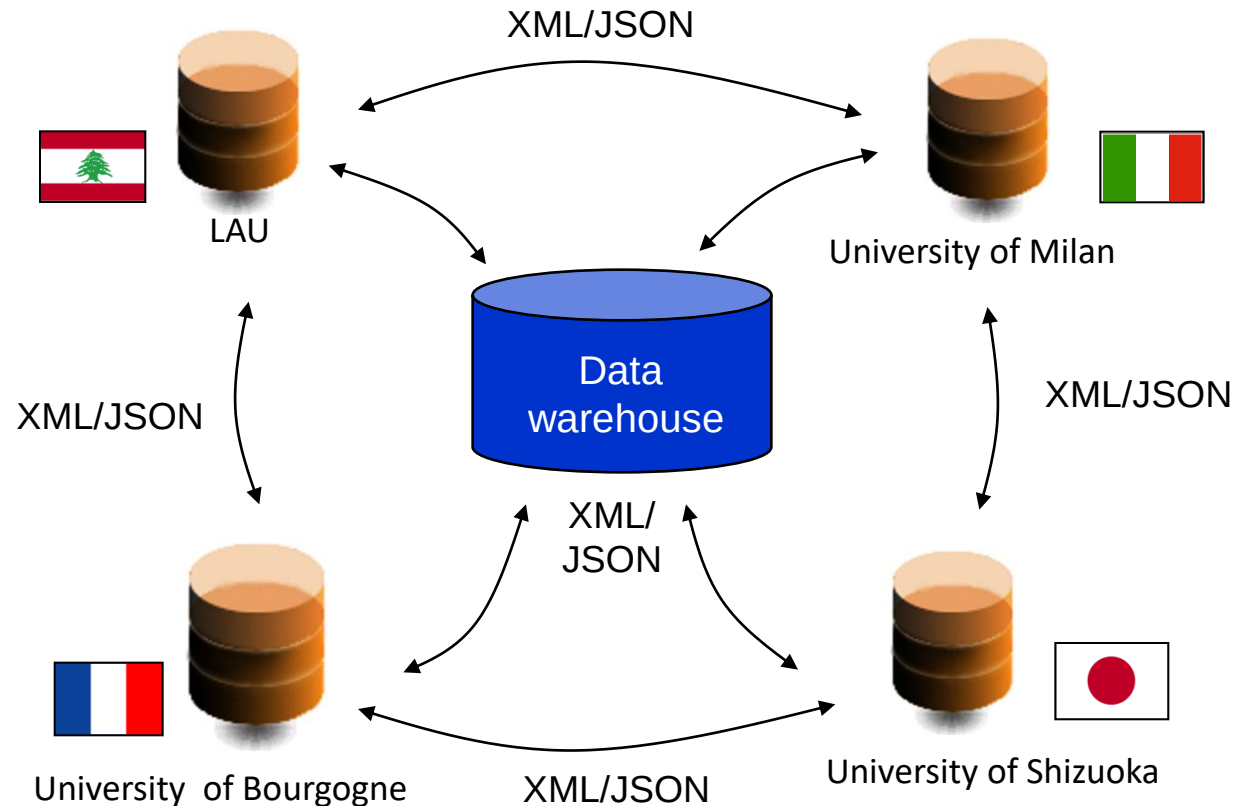
Low
Similarity



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1. Motivations

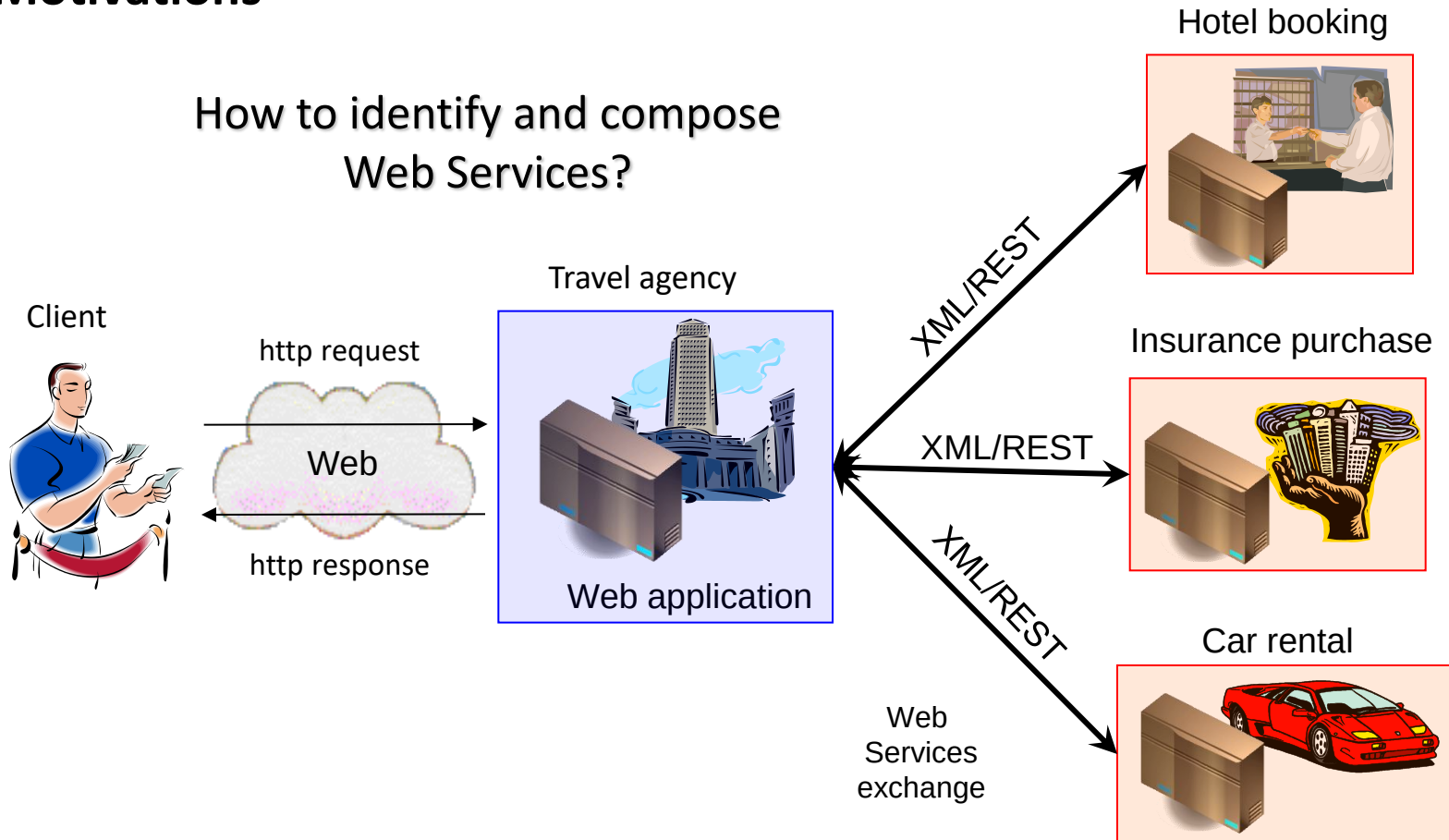
How to exchange
and integrate
information
between DBs?



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1. Motivations

How to identify and compose Web Services?



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1. Motivations

All these applications and others require some notion of
Semi-structured Similarity-based Processing

- Intelligent Web search and retrieval
- Data integration
- Service Composition
- Service performance
- Data warehousing
- Change management
- Storage and indexing
- Message translation
- ...

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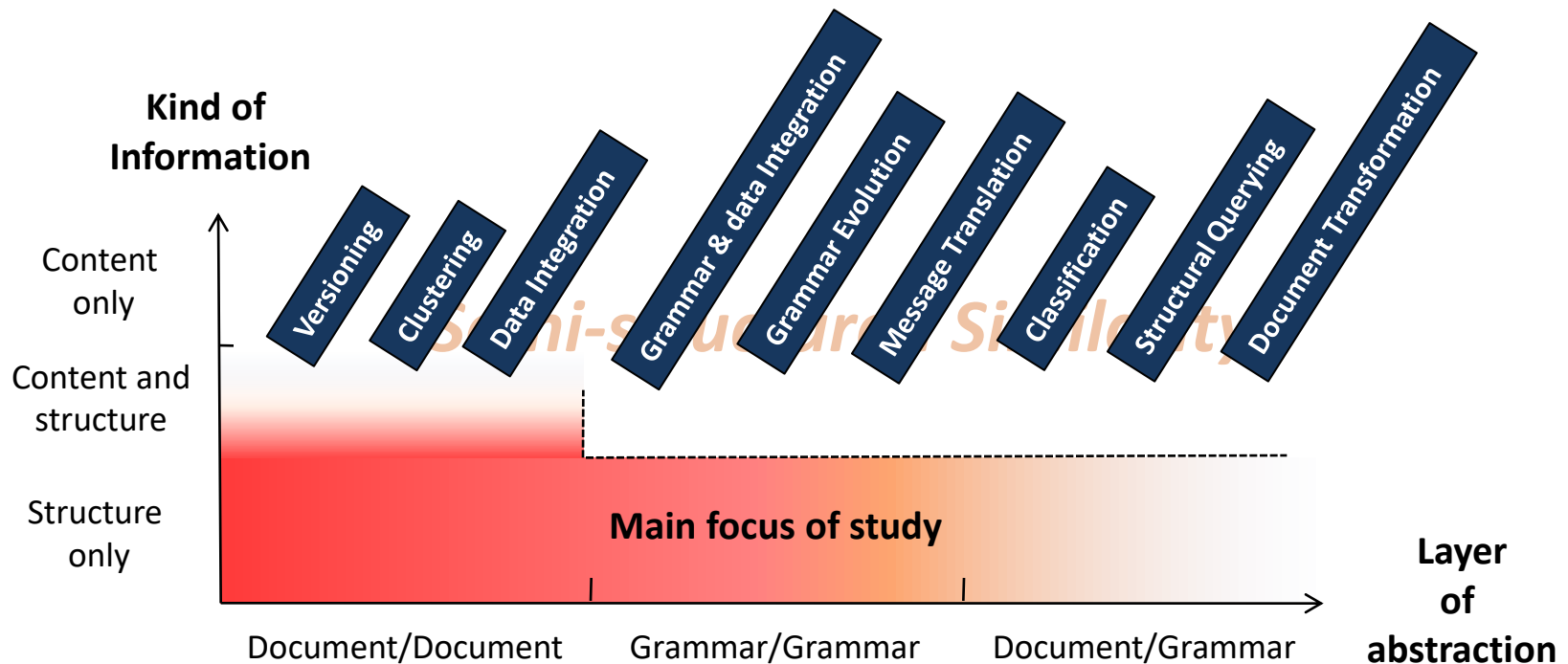
2. Formal Definition of Similarity

- **Inverse of a distance function (metric)**

- $\text{Sim}(S_1, S_2) \in [0, 1]$
- $\text{Sim}(S_1, S_2) = 1 \Leftrightarrow$ objects S_1 and S_2 are identical
- $\text{Sim}(S_1, S_2) = 0 \Leftrightarrow S_1$ and S_2 have no common characteristics
- Similarity increases with the commonality between S_1 and S_2 , and decreases with their difference
- Similarity and distance are inverse to each other
- $\text{Sim}(S_1, S_1) = 1 \Leftrightarrow$ **Reflexivity**
- $\text{Sim}(S_1, S_2) = \text{Sim}(S_2, S_1) \Leftrightarrow$ **Symmetry**
- $\text{Sim}(S_1, S_3) \geq \text{Sim}(S_1, S_2) \times \text{Sim}(S_2, S_3) \Leftrightarrow$ **Triangular inequality**
 - $\text{Dist}(S_1, S_3) \leq \text{Dist}(S_1, S_2) + \text{Dist}(S_2, S_3)$

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3. Main Categories of Similarity Approaches



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3. Main Categories of Similarity Approaches

- A range of algorithms for comparing semi-structured data have been proposed
 - Most of them exploit *CPM* and **AI techniques**
 - Rigorously structured data
 - Fine-grained
 - **Main target:** data structure
 - Others focus on extending conventional **IR methods**, e.g. the vector space model, so as to provide semi-structured similarity assessment
 - Loosely structured data
 - Coarse-grained
 - **Main target:** data content
 - Other diverse methods
 - Edge/Path Similarity, Fast Fourier Transform, Entropy...
 - **Usually approximate *Tree Edit Distance***

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4. Edit Distance based Approaches

- Making use of dynamic programming techniques for finding the *Edit Distance* between trees
 - Combinatorial Pattern Matching (CPM)

Finding the cheapest sequence of edit operations that can transform one tree into another

- Rigorously structured (*data-centric*) data
- Fine-grained
- **Main target:** data structure
- **Applications:** data warehousing and structural clustering

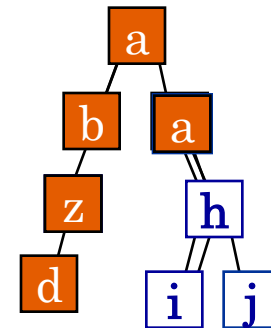
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4. Edit Distance based Approaches

- Algorithms can be distinguished following:
 - The set of edit operations allowed
 - *Insert node*: Insertion of inner/leaf nodes
 - *Delete node*: Deletion of inner/leaf nodes
 - *Update node*: Relabelling nodes
 - *Insert tree*
 - *Delete tree*
 - *Move tree*
 - The overall complexity and performance
 - $O(N^2 D^2)$
 - $O(N^2)$
 - $O(N \log(N))$



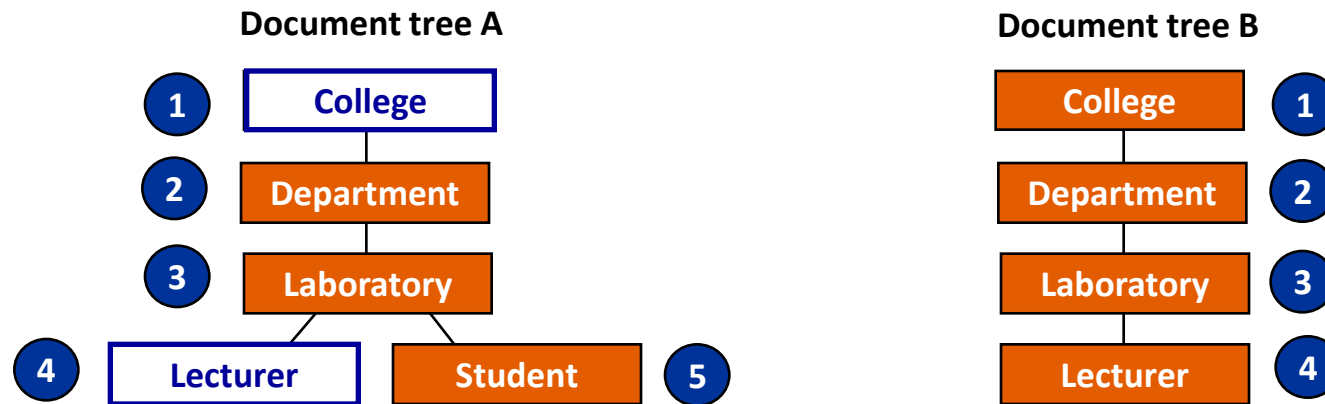
Sacrificing optimality



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4. Edit Distance based Approaches

Simple Example:



Edit script = $\text{Upd}(A[1], B[1]), \text{Upd}(A[4], B[4]), \text{Del}(A[5], A[3])$

$$\text{Dist}(A, B) = 3$$

$$\text{Sim} = 1 / 1 + \text{Dist}$$

$$\text{Sim}(A, B) = 0.25$$

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5. Information Retrieval based Approaches

- Documents are represented weighted vectors
 - Taking into account term occurrences and structure

**Comparing document vector representations
(cosine measure, Jaccard measure, Dice, ...)**

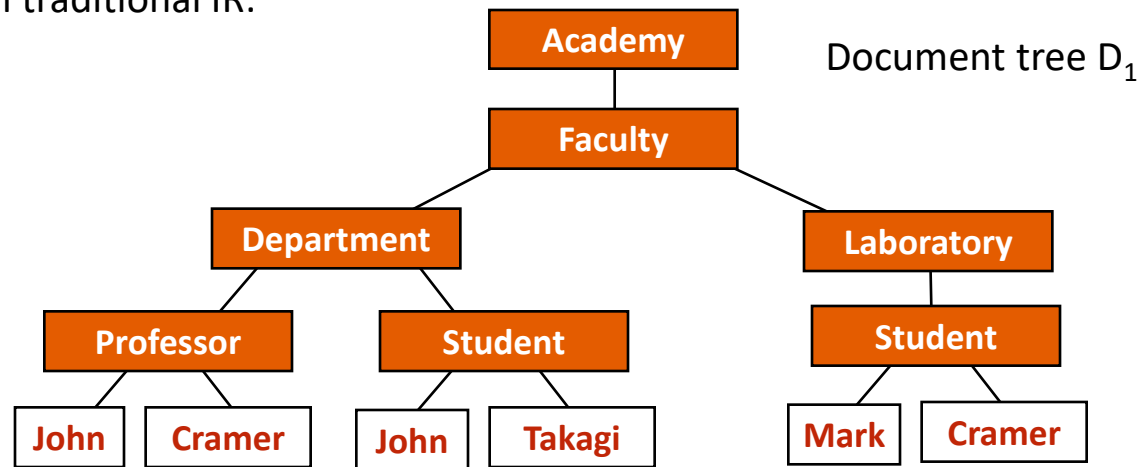
- Loosely structured (*document-centric*) data
- Coarse-grained
- **Main target:** data content
- **Application:** Fast and simple data querying

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5. Information Retrieval based Approaches

Simple Example:

↳ With traditional IR:



$$V_{D_1} = \begin{matrix} & \text{John} & \text{Cramer} & \text{Takagi} & \text{Mark} \\ (& 2 & , & 2 & , & 1 & , & 1 &) \end{matrix} \quad \Rightarrow \quad \text{TF weights}$$

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5. Information Retrieval based Approaches

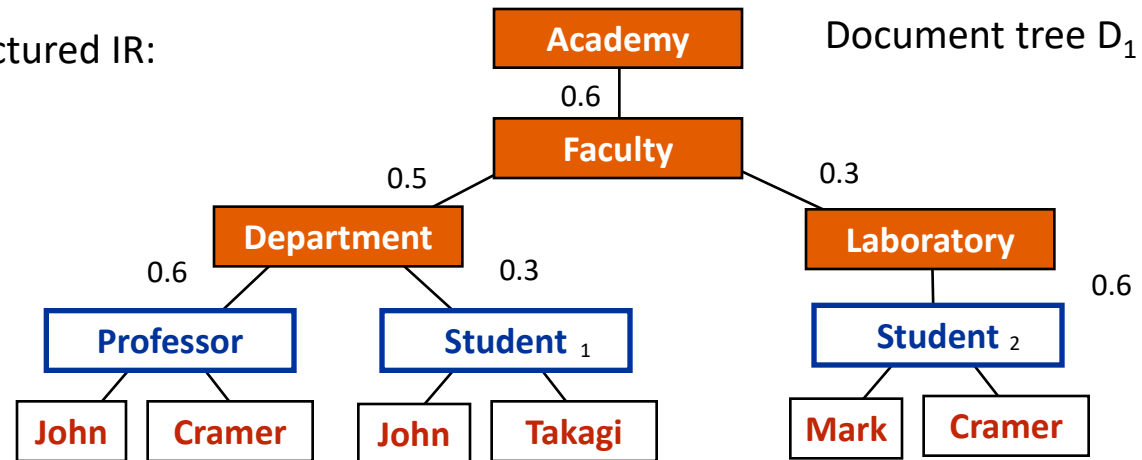
- **Traditional *IR* methods are not adequate with semi-structured data!**
 - With semi-structured data, content is distributed at different levels of the document tree
 - It is to be treated differently w.r.t. flat content to improve retrieval precision
 - **Down-weighting *TF-IDF* statistics w.r.t. structure**
 - Users in turn want to refer to document structure when searching for relevant information
 - Formulating ***Content-and-structure*** queries, in comparison with the ***content-only*** queries in conventional IR
 - Specifying search context (query scope)

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5. Information Retrieval based Approaches

Simple Example:

↳ With semi-structured IR:



	John	Cramer	Takagi	Mark	
$V_{\text{Professor}} =$	(1 ,	1 ,	0 ,	0)	TF scores
$V_{\text{Student1}} =$	(1 ,	0 ,	1 ,	0)	
$V_{\text{Department}} =$	(0.9 ,	0.6 ,	0.3 ,	0)	Down-weighted TF scores
$V_{\text{Faculty}} =$	(0.45 ,	0.48 ,	0.15 ,	0.18)	

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6. Other Approaches

- **Approximations of TED-based methods**
 - Tag similarity [Buttler 04]
 - Edge similarity [Lian et al., 04]
 - Path similarity [Rafiei et al., 06]
- **Extensions of IR-based methods**
 - Independent Component Analysis [Wang et al., 06]
 - Latent Semantic Kernel [Yang et al., 05] [Nayak et al. 08]
- **Others:**
 - Fast Fourier Transform (FFT) [Flesca et al., 02]
 - Leaf node clustering [Liang and Yokota, 05]
 - Entropy [Helmer 07]
 - Bayesian Networks [Leitao et al., 07]
 - Global similarity function [Nayak et al., 08] [Hwang and Ryu, 10]

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7. Current Issues

- Two main areas of study:
 - Enhancing **structural similarity**
 - Integrating **semantic similarity evaluation**



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7. Current Issues

7.1. Enhancing Structural Similarity

- Semi-structured documents tend to have many **optional and repeated elements**
- Such elements induce **recurring sub-trees** of similar or identical structures

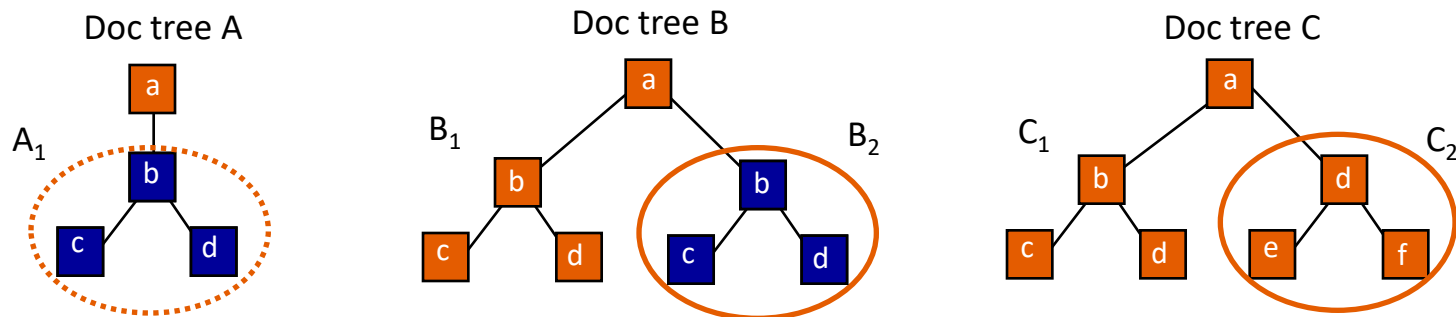
**Such repetitions/resemblances should be detected
to efficiently assess structural similarity**

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7. Current Issues

7.1. Enhancing Structural Similarity

- Identifying certain cases where existing edit distance approaches are inaccurate:
 - Similarity between document trees A/B (sub-trees A_1 and B_1) w.r.t. A/C**



$$\text{Sim}(A, B) = \text{Sim}(A, C)$$

?

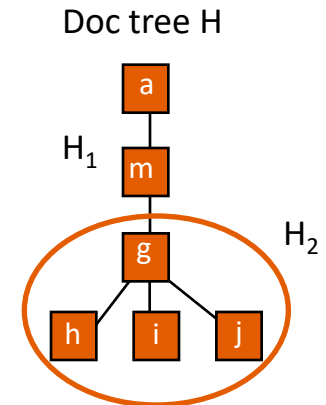
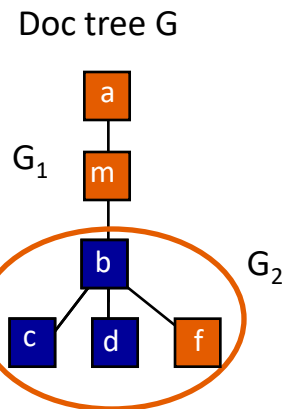
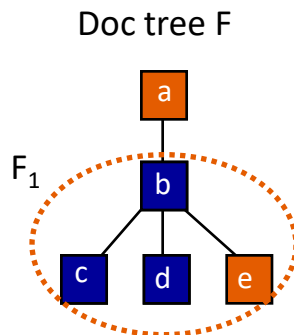
Repetition of similar sub-trees

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7. Current Issues

7.1. Enhancing Structural Similarity

- Identifying certain cases where existing edit distance approaches are inaccurate:
 - Similarity between document trees F and G (sub-trees F_1 and G_2) w.r.t. F/H**



$$\text{Sim}(F, G) = \text{Sim}(F, H) \quad ?$$

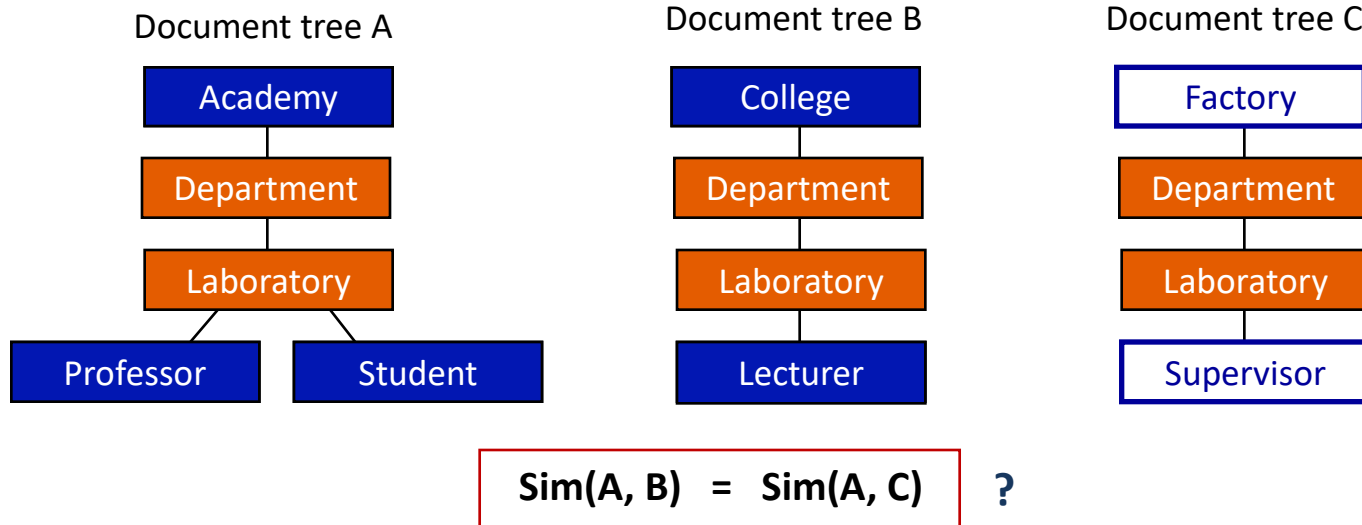
Structural similarity between sub-trees of different depths

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7. Current Issues

7.2. Integrating Semantic Similarity

- Most algorithms for comparing semi-structured data:
 - Focus on the structure of documents
 - Ignore the semantics involved



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7. Current Issues

- To sum up:

Problems

Detecting occurrences/repetitions of
structurally and **semantically** similar sub-trees



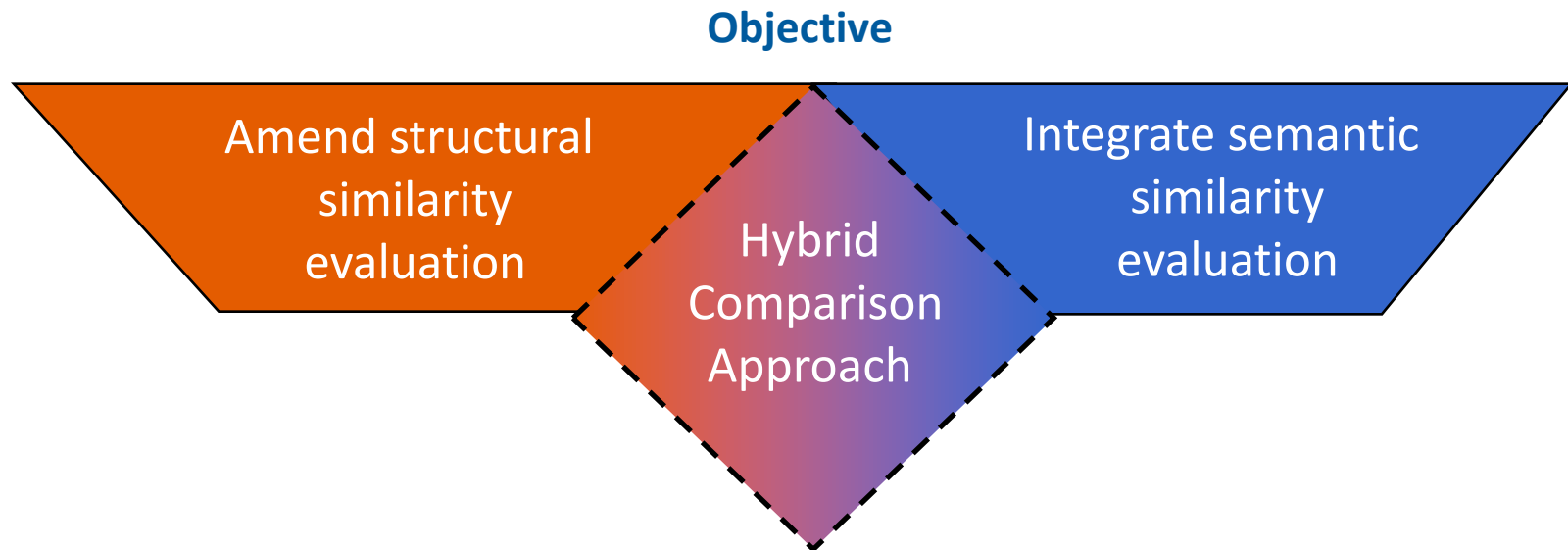
Improve comparison accuracy

Without drastically affecting complexity...

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7. Current Issues

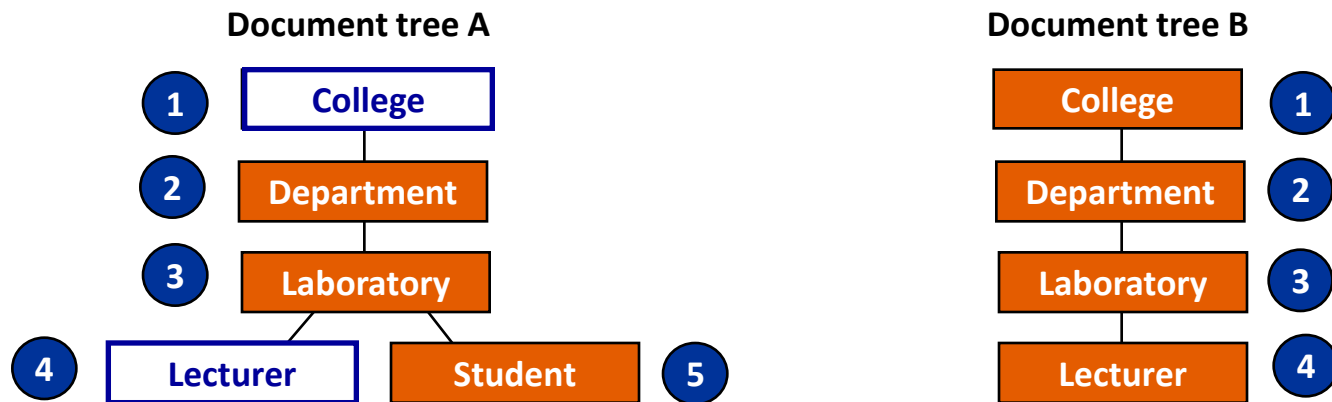
- To sum up:



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8. On-going Solutions

- **Structural Similarity**
 - Specialized Tree Edit Distance: Computing adapted **operations costs**



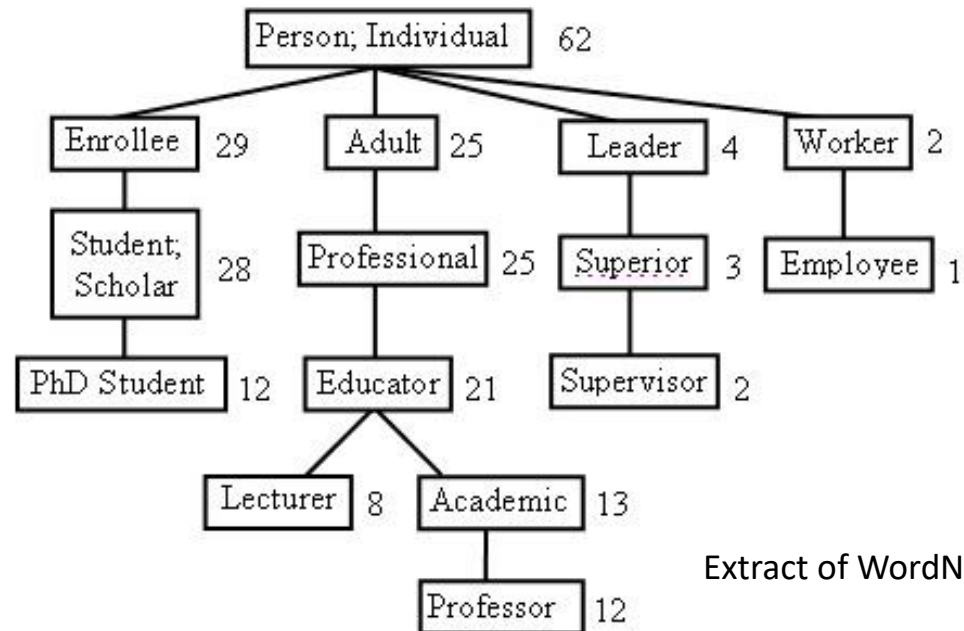
Edit script = Upd(A[1], B[1]), Upd(A[4], B[4]), Del(A[5], A[3])

$$\text{Dist}(A, B) = 3? \Rightarrow \text{Sim} = 1 / 1 + \text{Dist} \Rightarrow \text{Sim}(A, B) = 0.25?$$

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8. On-going Solutions

- **Semantic Similarity**
 - Making use of semantic references
 - Dictionaries
 - Thesauri
 - Taxonomies
 - Ontologies



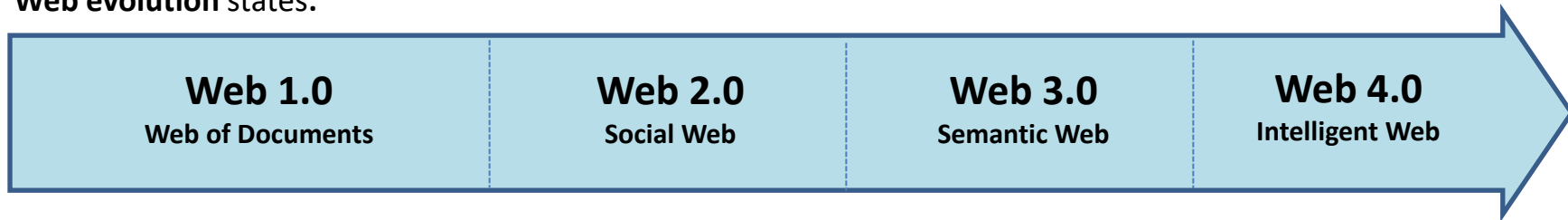
Extract of WordNet

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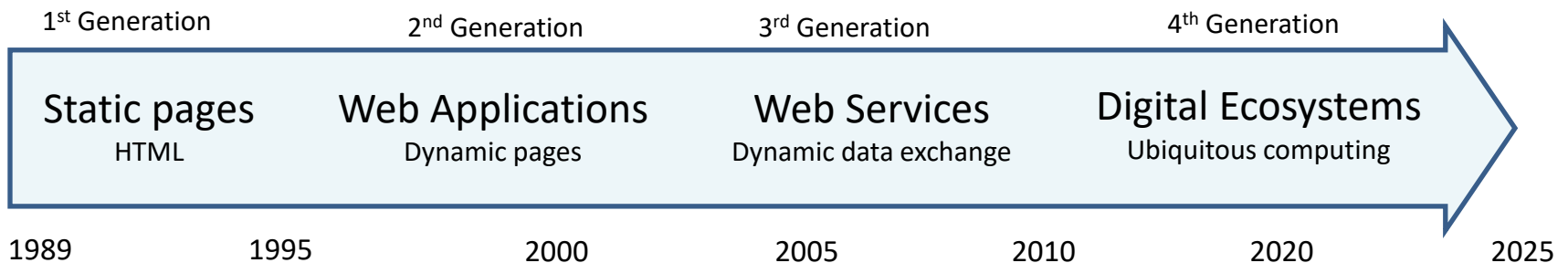
9. Toward an Intelligent Web

To put things back into perspective:

Web evolution states:



Recall **Web technologies evolution** (cf. Chapter 1):



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9. Toward an Intelligent Web

9.1. Web of Documents (Web 1.0)

- Genesis of the Web
 - Users access information in the form of HTML pages
 - Web documents
 - Information in Web documents is static
 - Document contents can only be modified manually
 - Web document contents are created by experts
 - Typical (non-expert) users are mere consumers of information



Web 1.0 is a mere forum for **publishing static** expert **information**

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9. Toward an Intelligent Web

9.2. Social Web (Web 2.0)

- Also known as **Collaborative Web**
 - Many users produce and consume contents simultaneously
 - **Blogs**: time stamped data entries
 - **Content communities**: sharing media contents between users
 - YouTube, Flickr, BookCrossing, etc.
 - **Social networking sites**: users connect by creating personal profiles, inviting friends, and sending e-mails and instant messages between each other
 - Facebook, MySpace, etc.
 - **Collaborative systems** (Wikis)
 - *Wiki* is derived from Hawaiian word: *Fast*
 - E.g., Wikipedia: ever more popular for content management

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9. Toward an Intelligent Web

9.2. Social Web (Web 2.0)

- Also known as **Collaborative Web**

Web 1.0	Web 2.0
Personal Web pages - Static contents	Blogs and social network profiles - Dynamic contents
Encyclopedia Britannica Online - Expert information only	Wikis - Expert and non-expert information
Content Publishing - Users are only consumers of information	Collaborative projects - Users create and consume information



Web 2.0 is a medium for **sharing dynamic user information**

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9. Toward an Intelligent Web

9.3. Semantic Web (Web 3.0)

- As the Web became increasingly interactive: Social Web (2.0)
 - where users were publishing data and interacting with each other
 - ⇒ Advent of a new era of distributed data management with double interaction:
 - i. Horizontal: **user-user**
 - ii. Vertical: **user-machine**
-  Need for more effective user-machine interactions
- **Tim Berners-Lee** introduces the vision of the **Semantic Web**:
 - Extending the Web by associating data with well-defined semantics (knowledge)
 - ⇒ Better enabling computers and people to interact and work in cooperation

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9. Toward an Intelligent Web

9.3. Semantic Web (Web 3.0)

What is **data semantics**? What is **knowledge**?

- To answer the question, we need to cover some basic concepts:
 - **Data:** the lowest abstraction of data representation with no meaning
 - Example: “2001” is considered as a number consisting of 4 digits
 - **Information:** The interpretation of the data by giving it well-defined meaning
 - Example “2001” is *“the year of announcement of the Semantic Web”*
 - **Meta-data:** A description about the data and/or information
 - Such as who gave the data/information (e.g., *Wikipedia*), when was the data/information given (e.g., *published in 2002*, etc.)

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9. Toward an Intelligent Web

9.3. Semantic Web (Web 3.0)

What is **data semantics**? What is **knowledge**?

- **Knowledge:** The combination of all known ***data***, ***information***, and ***meta-data*** concerning a given concept or fact, as well as the relations between them
 - Example: like knowing that “*the year of announcement of the Semantic Web*” is “2001”, following *Wikipedia* in an article *published in 2002*
- **Data Semantics:** The meaning of data, expressed in terms of its:
 - ***Information***
 - ***Meta-data***
 - Or at a higher level of abstraction and expressiveness: ***knowledge***

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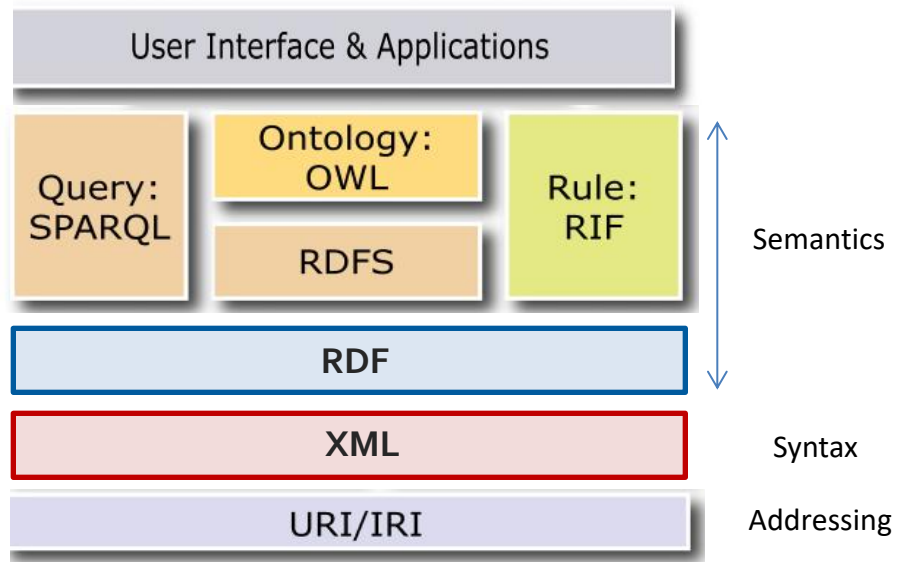
9. Toward an Intelligent Web

9.3. Semantic Web (Web 3.0)

- **Building blocks**

SW standards to describe data and semantics:

- XML: Extensible Markup Language
- RDF: Resource Description Framework
- RDFS: RDF Schema
- OWL: Ontology Web Language
- SPARQL: RDF/OWL query language



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9. Toward an Intelligent Web

9.3. Semantic Web (Web 3.0)

- To sum up, The Semantic Web vision promises to:
 - Give information well-defined meaning
 - ⇒ so that it becomes more easily accessible
 - by **human users** and **automated processes!**



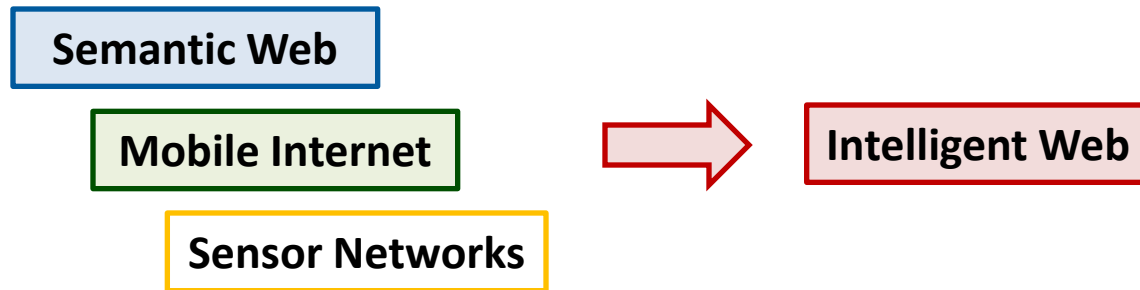
Web 3.0 is a platform for **sharing knowledge** between **humans** and **computers**

- **NOTE:** We are currently in the Semantic Web era
 - Examples: Google, Facebook, Flickr, Wikis, etc. are all becoming semantically sensitive
 - Allowing semantic data annotations, search, and manipulation

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9. Toward an Intelligent Web

9.4. Intelligent Web (Web 4.0)



- **Intelligent Web** (IW), also known as the **Internet of Things** (IoT) vision
 - Extension of the Semantic Web where:
 - Objects of the real world (with added digital components)
 - Objects of the virtual world (software agents on the Web)
 - ⇒ Are connected and interact at anytime/anyplace, using:
 - **Collective knowledge**
 - **Ubiquitous computing**

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9. Toward an Intelligent Web

9.4. *Intelligent Web (Web 4.0)*

- **Collective knowledge:** Knowledge acquired from a distributed pool of sources
 - The combination of all known data, information, and meta-data concerning a given (set of) concept(s), fact(s), user(s), or processes (s)
 - As well as the semantic relations between them

Intelligent Web terminals are expected to automatically handle CK



To improve information management and services

- **Challenges:**
 - Knowledge Evaluation and Verification
 - Knowledge Versioning and Preservation
 - Knowledge Extrapolation

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9. Toward an Intelligent Web

9.4. *Intelligent Web (Web 4.0)*

- **Ubiquitous Connectivity:** Always-on connectivity, based on:
 - Developments in network technologies:
 - RFID (Radio Frequency Identification)
 - Short-range wireless technologies (WI-FI)
 - Sensor networks
 - Enhancements in network addressing techniques:
 - IPv6 to expand address space
 - IRI (Internationalized Resource Identifier)

Reaching more objects in the physical/virtual worlds



Connecting them together through the Internet

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9. Toward an Intelligent Web

9.4. Intelligent Web (Web 4.0)

- Extending the concept of cooperation from:
 - **User-User** and **User-Machine** with the Semantic Web
 - ⇒ To **Machine-Machine** interaction
- Machines do not simply interact with humans
 - **Machines interact with each other!**
- Interaction is not only based on a simple exchange of *data*
 - But rather on an exchange of semantically meaningful *information*
 - Exchange of knowledge well understood by machines

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9. Toward an Intelligent Web

9.4. *Intelligent Web (Web 4.0)*

Web 4.0 promotes the concept of **Digital Web Ecosystems**

↳ Human and software agents collaborate, interact, compete, and evolve autonomously
⇒ In order to automatically solve complex and dynamic problems

- **NOTE:** We are not there yet! We are currently in the Semantic Web era
 - Very hot research topic!

↳ Building on Semi-structured, Semantic, and Linked data processing

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Questions?



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Personal assignment

- **Learning to perform XML and JSON parsing:**
 - **SAX** - Megginson D. et al. The Simple API for XML <http://www.megginson.com/SAX/>
 - **Xerces XML Parser.** Xerces Apache Foundation. Available from <http://xerces.apache.org/>
 - **The Expat XML Parser.** Expat SourceForge.NET. Available from <http://expat.sourceforge.net/>
 - **XBIS** - SourceForge.NET. XML Binary Information Set (XBIS). Available from: <http://xbis.sourceforge.net/>
 - <https://www.geeksforgeeks.org/parse-json-java/>
 - <https://learn.microsoft.com/en-us/dotnet/standard/serialization/system-text-json/how-to?pivots=dotnet-8-0>
 - <https://linuxhint.com/parse-json-data-cpp/>
 - Many other XML/JSON parsers are embedded in programming environments...

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Exercises and Tutorials

- Provided by the instructor during lecture sessions



Presentation assignment



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Main References

- Joe Tekli, Gilbert Tekli, Richard Chbeir: Combining offline and on-the-fly disambiguation to perform semantic-aware XML querying. *Comput. Sci. Inf. Syst.*, 2023, 20(1): 423-457
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- Sarkis Sarkissian, Joe Tekli: Unsupervised Topical Organization of Documents using Corpus-based Text Analysis. *MEDES 2021*: 87-94, 2021
- Joe Tekli, Richard Chbeir, Agma J. M. Traina, Caetano Traina Jr., Renato Fileto: Approximate XML structure validation based on document-grammar tree similarity. *Inf. Sci.*, 2015 295: 258-302
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Various other scientific papers and publications...